

(4) [10 points]

Assume that printLL prints all items in a linked list in order on one line. What will be printed by the following program?

// printLL function is defined here.

```
struct IntItem {
    int value;
    IntItem *next;
    IntItem (int v, IntItem *n) { value = v; next = n; }
};

IntItem* mystery1 (int start, int end) {
    if(start < end) return new IntItem(start, mystery1(start+1,end));
    else return nullptr;
}

IntItem* mystery2 (IntItem* head, IntItem* prev) {
    if(head != nullptr) return mystery2(head->next, new IntItem(head->value, prev));
    else return prev;
}

int main() {
    IntItem* p1 = mystery1(0,5);
    printLL(p1);
    IntItem* p2 = mystery2(p1, nullptr);
    printLL(p2);
    return 0;
}
```

Outputs:

0 1 2 3 4
4 3 2 1 0

